AWS Storage Lesson

Skills you will learn

Cloud Computing: Storage & Databases

Lesson Overview

In this lesson, we will explore the different storage solutions AWS offers. We will cover storage types, specific services like Amazon EBS, S3, and EFS, and AWS migration tools for transferring data to the cloud.

Learning Objectives

By the end of this lesson, you will be able to:

1. Understand the different AWS storage solutions.
2. Explore Amazon S3 and its features.
3. Identify AWS migration tools for data transfer.

AWS Storage Overview

Cloud storage is essential for SaaS applications like iCloud, Google Drive, and Dropbox. AWS provides three types of storage:

1. **Block Storage** → Amazon EBS
2. **File Storage** → Amazon EFS, FSx for Lustre
3. **Object Storage** → Amazon S3

Amazon Elastic Block Store (EBS)

Amazon EBS provides durable, detachable block-level storage for EC2 instances. It supports various volume types tailored for performance and cost-efficiency.

EBS Volume Types:

- **General Purpose SSD (gp3, gp2)** → Balanced performance.

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- **Provisioned IOPS SSD (io2, io1)** → High-performance, low-latency storage.
- **Throughput Optimized HDD (st1)** → Optimized for frequently accessed workloads.
- **Cold HDD (sc1)** → Cost-efficient for infrequently accessed data.

Amazon Simple Storage Service (S3)

Amazon S3 is an object storage solution designed for high availability, security, and cost-efficiency.

S3 Use Cases:

- Backup and Restore
- Static Website Hosting
- Media Storage and Streaming
- Data Archiving
- Analytics and Data Lakes

S3 Storage Classes:

Storage Class	Description
S3 Standard	General-purpose storage
S3 Standard-IA	Infrequent access, lower cost
S3 One Zone-IA	Infrequent access, stored in a single AZ
S3 Glacier Instant Retrieval	Archive storage with millisecond retrieval
S3 Glacier Flexible Retrieval	Archive storage, retrieval in minutes to hours
S3 Glacier Deep Archive	Long-term storage, retrieval in up to 12 hours

AWS Intelligent Tiering automatically moves data between storage classes based on usage patterns.

Amazon S3 Security & Features

Access Control

- **S3 Bucket Policies** → JSON-based access control rules.
- **S3 Access Points** → Custom access control per bucket.

Additional Features

- **Versioning** → Retains multiple object versions.
- **Object Lock** → Prevents accidental deletions.
- **Lifecycle Policies** → Moves/deletes objects based on age.

- **Event Notifications** → Triggers alerts for S3 events.
- **Multipart Upload** → Uploads large files in parts.
- **Transfer Acceleration** → Faster global data transfers.

Amazon Elastic File System (EFS)

Amazon EFS is a scalable, elastic file storage system that supports multiple EC2 instances simultaneously.

EFS Features:

- **Grows and shrinks dynamically.**
- **Pay-as-you-go pricing.**
- **Supports burst throughput for high performance.**

AWS Data Migration Tools

AWS provides tools for migrating data between on-premises systems and AWS storage services.

- **AWS DataSync** → Transfers data to S3, EFS, or FSx for Windows.
- **AWS Transfer Family** → Secure FTP transfers to/from S3.
- **AWS Storage Gateway** → Extends on-prem storage to AWS.
- **AWS Snow Family** → Physical device-based data transfer for large volumes.

Completion Criteria

- Understand different AWS storage solutions.
- Identify the use cases for Amazon S3, EBS, and EFS.
- Learn AWS data migration tools.

🚀 **Great job! You've successfully completed AWS Storage Services!** 🎉

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit